

S07 required architecture decision record brief and blank template

Required analysis · full P03 implementation optional · one final S07 PDF

Teaching material FINAL_LOCAL v1.2.1 · local Phase4 closure · native and external acceptance pending.

Choose an existing sanitised semester resource or an explicitly provisional registration/booking scenario. The 300-500-word length plus a small repeat table is a teaching target, not a canonical word-count requirement. Finish after the content meeting before the later configured deadline. An incomplete ADR remains incomplete even when optional P03 code exists.

Eight linked parts

1. **Context and invariant (adr_context).** Name the scenario, stakeholders and facts that must agree. State whether the scenario is selected or provisional.
2. **Resource identity and representation (adr_resources).** Explain how the resource is identified and which fields are public. Distinguish public representation from internal ORM names.
3. **Alternatives (adr_alternatives).** Compare at least two viable designs. What does each make easy, uncertain or costly?
4. **Decision (adr_decision).** Choose one and justify it against your context, not a universal URL slogan.
5. **Transaction scope (adr_transaction).** Name the owner, read/write set, constraints and error policy. Identify what rolls back together and what remains outside. Do not infer external reversibility from await.
6. **Repeat table (adr_retry_table).** Complete the eight rows below from the P03 contract. Explain effect versus response and uncertain-response behaviour.
7. **Evidence (adr_evidence).** Name exact source paths/sections and any actual request/model witness separately. Source reading is SOURCE_REASONING and is allowed here.
8. **Consequences (adr_consequences).** Explain costs, assumptions, untested concurrency/crash policies and a concrete review trigger.

Blank eight row repeat table

Use contracts/P03.md for authoritative expected categories. This template intentionally leaves your decision and response/effect explanation blank. A hypothetical lost response is a thought experiment unless you actually observed loss.

Case	Method/path	Intended effect	Expected response category	Evidence class/source	Uncertainty
Collection read	Fill	Fill	Fill	Fill	Fill
First member PUT	Fill	Fill	Fill	Fill	Fill
Identical member PUT	Fill	Fill	Fill	Fill	Fill

Case	Method/path	Intended effect	Expected response category	Evidence class/source	Uncertainty
PUT changing only ticketType	Fill	Fill	Fill	Fill	Fill
First member DELETE	Fill	Fill	Fill	Fill	Fill
Repeated DELETE when absent	Fill	Fill	Fill	Fill	Fill
Invalid query or body	Fill	Fill	Fill	Fill	Fill
Missing parent	Fill	Fill	Fill	Fill	Fill

Scope that your decision must preserve

P02 booking uses a supplied POST and is not automatically idempotent. P03 registration identity is (sessionId, attendeeId) and its member PUT/DELETE express a target state. Repeated idempotent effects need not return identical status codes. P02 Location names a booking but has no supplied GET member route. Do not invent that route or an automatic retry policy.

Sequential rollback evidence does not qualify concurrent writers, crash durability or external effects. A broad UniqueConstraintError class alone cannot identify its originating constraint. State these limits rather than implementing a new isolation/retry policy in the one-file P02 task.

Your own writing prompts

Context: _____

Identity and representation: _____

Alternative A and B, with trade-offs: _____

Chosen decision and reason: _____

Transaction owner, read/write set and error provenance: _____

Repeat table reasoning: _____

Evidence references and classes: _____

Consequences, uncertainty and review trigger: _____

Include this completed analysis in the same final S07 PDF. Do not create a second C07 Assignment or publish the work on GitHub. Missing optional full P01/P03 code must not reduce the standard maximum mark.